txt2tex Cheatsheet

Write math like you're at a whiteboard. Get LaTeX you can hand in.

https://github.com/jmf-pobox/txt2tex

Document Structure

`=== Title ===`	`\section *{Title}`
`* Solution N **`	Bold heading with spacing
`(a) Text`	Part label
`TEXT: paragraph`	Prose block (smart formula detection)
`LATEX: \cmd`	Raw LaTeX passthrough
`PAGEBREAK:`	Page break

`TITLE: ...`	Document title
`AUTHOR: ...`	Author name
`DATE: ...`	Date string
`BIBLIOGRAPHY: f.bib`	BibTeX file
`[cite key]`	`\citep {key}`

Logic

`p land q`	$p \land q$
`p lor q`	$p \lor q$
`lnot p`	$\neg p$
`p => q`	$p \Rightarrow q$
`p <=> q`	$p \Leftrightarrow q$
`forall x : S | P`	$\forall x : S \bullet P$
`exists x : S | P`	$\exists x : S \bullet P$
`exists_1 x : S | P`	$\exists_1 x : S \bullet P$
`lambda x : S | E`	$\lambda x : S \bullet E$
`mu x : S | P`	$\mu x : S \bullet P$

Note: Use `land`, `lor`, `lnot`. English `and`/`or`/`not` are not supported.

Sets & Types

`x elem S`	$x \in S$
`x notin S`	$x \notin S$
`A subset B`	$A \subseteq B$
`A union B`	$A \cup B$
`A inter B`	$A \cap B$
`A setminus B`	$A \setminus B$
`power S`	$\mathbb{P}\, S$
`F S`	$\mathbb{F}\, S$
`{x : S | P}`	Set comprehension
`A cross B`	$A \times B$
`N / Z / N1`	$\mathbb{N}$ / $\mathbb{Z}$ / $\mathbb{N}_1$
`n..m`	$n \mathinner{..} m$

Relations

`A <-> B`	$A \leftrightarrow B$
`x |-> y`	$(x \mapsto y)$
`dom R / ran R`	$\mathrm{dom}\, R$ / $\mathrm{ran}\, R$
`A dres R`	$A \lhd R$
`A dsub R`	$A \lhd\!\!\!\cdot\, R$
`R rres B`	$R \rhd B$
`R rsub B`	$R \rhd\!\!\!\cdot\, B$
`R oplus S`	$R \oplus S$ (override)
`R o9 S`	$R \fatsemi S$ (forward comp.)
`R comp S`	$R \fatsemi S$ (backward comp.)

Functions

`A pfun B`	$A \mathrel{+\!\!\!\rightarrow} B$
`A fun B`	$A \longrightarrow B$ (total)
`A pinj B`	$A \rightarrowtail\!\!\!\!\!+ B$
`A inj B`	$A \rightarrowtail B$
`A surj B`	$A \twoheadrightarrow B$
`A bij B`	$A \rightarrowtail\!\!\!\!\!\twoheadrightarrow B$
`A ffun B`	$A \mathrel{+\!\!\!\twoheadrightarrow} B$ (finite)
`f(x)`	Function application

Sequences

`<> / <a, b>`	$\langle\rangle$ / $\langle a, b\rangle$
`s ^ t`	$s \frown t$ (concatenation)
`seq S / seq1 S`	$\mathrm{seq}\, S$ / $\mathrm{seq}_1\, S$
`# s`	$\#s$ (length)

Z Notation Blocks

`given A, B`	Given types: $[A, B]$
`T ::= a | b`	Free type
`name == expr`	Abbreviation

`axdef` / `schema Name` / `gendef [X]`

`declarations`	Variable declarations
`where`	Separator
`predicates`	Constraining predicates
`end`	Close the block

Proof Tree Syntax

indent	Premises indented under conclusion
`[rule]`	Justification label
`[rule [n]]`	Discharge assumption n
`[n] expr`	Boxed assumption labelled n
`::`	Sibling premises

Rules: `land intro`, `land elim 1/2`, `lor intro 1/2`, `lor elim`, `=> intro/elim`, `<=> intro/elim`, `lnot intro/elim`, `forall intro/elim`, `exists intro/elim`.

Usage

`txt2tex f.txt`	→ f.tex + f.pdf
`txt2tex f.txt --tex-only`	→ f.tex only
`txt2tex -i`	Interactive REPL
`txt2tex --check-env`	Verify dependencies

txt2tex Proofs — By Example

Each example shows what you type and what txt2tex renders.

How Proof Trees Work

In txt2tex, proof trees are written **bottom-up**: the conclusion is the first line, and its premises are indented below it. Think of it as reading the inference rule from conclusion to justification:

conclusion	First line (least indented)
premise	Indented = supports the line above
`[rule]`	Justification in brackets
`[n] expr`	Boxed assumption (labelled n)
`[rule [n]]`	Discharge assumption n
`::`	Marks sibling premises

Deeper indentation = deeper in the tree. Two premises at the same indent level are siblings that jointly support their parent.

Truth Table Prop. Logic

You write:

```
TRUTH TABLE:
p | q | p => q
T | T | T
T | F | F
F | T | T
F | F | T
```

You get:

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Equivalence Chain Prop. Logic

You write:

```
EQUIV:
lnot (p land q)
lnot p lor lnot q  [De Morgan]
```

You get:

$$\begin{aligned} &\neg\,(p \wedge q) \\ &\Leftrightarrow \neg\, p \vee \neg\, q \qquad \text{[De Morgan]} \end{aligned}$$

`EQUIV:` joins steps with $\Leftrightarrow$. Use for *predicate* equivalences.

Modus Ponens Proof Trees

You write:

```
PROOF:
  q [=> elim]
    p => q
    p
```

You get:

$$\frac{p \Rightarrow q \quad p}{q}\ \Rightarrow\text{-elim}$$

$\wedge$-intro Proof Trees

You write:

```
PROOF:
  p land q [land intro]
    p
    q
```

You get:

$$\frac{p \quad q}{p \wedge q}\ \wedge\text{-intro}$$

$\Rightarrow$-intro with discharge Proof Trees

You write:

```
PROOF:
  p => p [=> intro [1]]
    [1] p
```

You get:

$$\frac{\ulcorner p \urcorner^{[1]}}{p \Rightarrow p}\ \Rightarrow\text{-intro}^{[1]}$$

`[1] p` = boxed assumption. `[=> intro [1]]` discharges it.

$\vee$-elim / case analysis Proof Trees

You write:

```
PROOF:
  r [lor elim]
    p lor q
    :: p => r
      [1] p
        r [from p]
    :: q => r
      [2] q
        r [from q]
```

You get:

$$\frac{p \vee q \quad \dfrac{\ulcorner p \urcorner^{[1]}}{r} \quad \dfrac{\ulcorner q \urcorner^{[2]}}{r}}{r}\ \vee\text{-elim}$$

`::` = sibling premises (case arms).

Equality Chain Induction

You write:

```
EQUAL:
0 + 0
0       [0+0 = 0]
length(<>)
       [length(<>) = 0]
```

You get:

$$\begin{aligned} &0 + 0 \\ &= 0 && [0{+}0 = 0] \\ &= \mathit{length}(\langle\rangle) && [\mathrm{length}(\langle\rangle) = 0] \end{aligned}$$

`EQUAL:` = expression chains (=). Use for $\mathbb{N}$/sets/sequences.